

7.6-1

①

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1) If the curve is given in Cartesian form $\begin{cases} y = f(x) \\ \text{or } x = f(y) \end{cases}$

$$S = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{or} \quad S = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

2) If the curve is given in Parametric form $\begin{cases} x = f(t) \\ y = f(t) \end{cases}$

$$S = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

3) If the curve is given in Polar form $\begin{cases} r = f(\theta) \\ \text{or } \theta = f(r) \end{cases}$

$$S = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \quad \text{or} \quad S = \int_{r_1}^{r_2} \sqrt{1 + \left(r \frac{d\theta}{dr}\right)^2} dr$$

Ex 7.6

Q1 $y^2 = 4ax$ — (1)

(2)

$3y = 8x$ — (2)

$y = \frac{8x}{3}$

Solving (1) & (2) to find pt. of X.

$$\left(\frac{8x}{3}\right)^2 = 4ax$$

$$\frac{64x^2}{9} = 4ax \Rightarrow 64x^2 = 36ax \Rightarrow 64x^2 - 36ax = 0$$

$$4x(16x - 9a) = 0$$

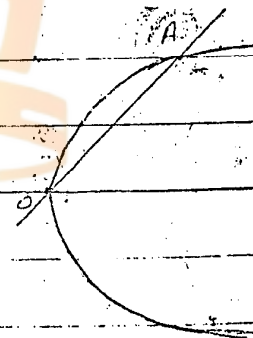
$$4x = 0, \quad 16x - 9a = 0$$

$$x = 0, \quad x = \frac{9a}{16} \quad \text{Put in (2)}$$

$$\text{when } x = 0 \text{ in (2) } y = 0$$

$$\text{when } x = \frac{9a}{16}, \quad 3y = \frac{8(9a)}{16}$$

$$3y = \frac{9a}{2} \Rightarrow y = \frac{3a}{2}$$



7.6-2

Hence pt of xn are $(0,0), (\frac{9a}{16}, \frac{3a}{2})$

$$S = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$y^2 = 4ax$$

$$= \int_0^{\frac{9a}{16}} \sqrt{\frac{x+a}{x}} dx$$

$$2y \frac{dy}{dx} = 4a$$

$$\frac{dy}{dx} = \frac{4a}{2y} = \frac{2a}{y}$$

$$\text{Put } x = a \tan^2 \theta$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{4a^2}{y^2} = \frac{4a^2}{4ax}$$

$$dx = a 2 \tan \theta \sec^2 \theta d\theta$$

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{a}{x}$$

$$\int_0^{\frac{9a}{16}} \sqrt{\frac{x+a}{x}} dx = \int \frac{a \tan^2 \theta + a}{a \tan^2 \theta} 2a \tan \theta \sec^2 \theta d\theta = \frac{x+a}{x}$$

$$= \int \frac{a(\tan^2 \theta + 1)}{a \tan^2 \theta} 2a \tan \theta \sec^2 \theta d\theta$$

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$$= \int \frac{\sec^2 \theta}{\tan^2 \theta} 2a \tan \theta \sec^2 \theta d\theta$$

$$= \int \frac{\sec \theta}{\tan \theta} 2a \tan \theta \sec^2 \theta d\theta = 2a \int \sec^3 \theta d\theta \quad \text{--- (3)}$$

$$\text{Consider } I = \int \sec^3 \theta d\theta = \int \sec^2 \theta \sec \theta d\theta \quad \text{IBP}$$

$$= \sec \theta \tan \theta - \int (\sec \theta \tan \theta) \tan \theta d\theta$$

$$= \sec \theta \tan \theta - \int \sec \theta \tan^2 \theta d\theta$$

7.6-3

$$I = \sec \theta \tan \theta - \int \sec \theta (\sec^2 \theta - 1) d\theta$$

$$= \sec \theta \tan \theta - \int \sec^3 \theta d\theta + \int \sec \theta d\theta$$

$$2I = \sec \theta \tan \theta + \int \sec \theta d\theta$$

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$$I = \frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta|$$

$$\int \sec^3 \theta d\theta = \frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| \quad \text{--- (4)}$$

$$\text{Using (4) in (3)} \quad \int_0^{\frac{9a}{16}} \frac{\sqrt{x+a}}{x} = \frac{2a}{\cancel{2}} \left[\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| \right]_0^{\frac{9a}{16}}$$

$$= a \left[\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta| \right]_0^{\frac{9a}{16}}$$

$$= a \left[\frac{\sqrt{x+a}}{a} \sqrt{\frac{x}{a}} + \ln \left| \frac{\sqrt{x+a}}{a} + \sqrt{\frac{x}{a}} \right| \right]_0^{\frac{9a}{16}} \quad \begin{aligned} \because x &= a \tan^2 \theta \\ \frac{x}{a} &= \tan^2 \theta \\ \sqrt{\frac{x}{a}} &= \tan \theta \end{aligned}$$

$$= a \left[\frac{\frac{9a}{16} + a}{a} \sqrt{\frac{9a}{16}} + \ln \left| \frac{\frac{9a}{16} + a}{a} + \sqrt{\frac{9a}{16}} \right| \right] \quad \begin{aligned} 1 + \tan^2 \theta &= \sec^2 \theta \\ 1 + \frac{x}{a} &= \sec^2 \theta \end{aligned}$$

$$= a \left[\frac{(\frac{9}{16} + 1)x}{x} \sqrt{\frac{9x}{16x}} + \ln \left| \frac{(\frac{9}{16} + 1)x}{x} + \sqrt{\frac{9x}{16x}} \right| \right] \quad \sqrt{a+x} = \sec \theta$$

$$= a \left[\sqrt{\frac{25}{16}} \sqrt{\frac{9}{16}} + \ln \left| \sqrt{\frac{25}{16}} + \sqrt{\frac{9}{16}} \right| \right]$$

$$= a \left[\frac{5}{4} \cdot \frac{3}{4} + \ln \left| \frac{5}{4} + \frac{3}{4} \right| \right] = a \left[\frac{15}{16} + \ln \left| \frac{8}{4} \right| \right]$$

$$= a \left[\frac{15}{16} + \ln 2 \right] \text{Ans.}$$

7.6-4

Easy Method of Q1.

$$y^2 = 4ax \quad \text{--- ①}$$

$$3y = 8x \quad \text{--- ②}$$

Solving ① & ② to find Pt of X_n .

$$\left(\frac{8x}{3}\right)^2 = 4ax \Rightarrow 64x^2 = 36ax$$

$$64x^2 - 36ax = 0 \Rightarrow 4x(16x - 9a) = 0 \Rightarrow x = 0, \frac{9a}{16}$$

Putting values of x in ② we get when $x=0, y=0$

$$\text{Pt. of } X_n = \left(0, 0\right) \left(\frac{9a}{16}, \frac{3a}{2}\right) \quad \text{when } x = \frac{9a}{16} \quad y = \frac{8(9a)}{3 \cdot 16} = \frac{3a}{2}$$

$$S' = \int_0^{\frac{3a}{2}} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$S' = \int_0^{\frac{3a}{2}} \sqrt{\frac{4a^2 + y^2}{4a^2}} dy$$

$$= \frac{1}{2a} \int_0^{\frac{3a}{2}} \sqrt{(2a)^2 + y^2} dy$$

$$= \frac{1}{2a} \left[\frac{y}{2} \sqrt{4a^2 + y^2} + \frac{4a^2}{2} \ln \left(\frac{y + \sqrt{4a^2 + y^2}}{2a} \right) \right]_0^{\frac{3a}{2}}$$

$$= \frac{1}{2a} \left[\frac{3a}{4} \sqrt{4a^2 + \frac{9a^2}{4}} + \frac{4a^2}{2} \ln \left(\frac{\frac{3a}{2} + \sqrt{4a^2 + \frac{9a^2}{4}}}{2a} \right) \right]$$

$$= \frac{1}{2a} \left[\frac{3a}{4} \sqrt{\frac{25a^2}{4}} + 2a^2 \ln \left(\frac{\frac{3a}{2} + \sqrt{\frac{25a^2}{4}}}{2a} \right) \right]$$

$$= \sec \theta \tan \theta - \ln | \sec \theta + \tan \theta |$$

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$$y^2 = 4ax$$

$$2y \frac{dy}{dx} = 4a$$

$$\frac{dy}{dx} = \frac{4a}{2y}$$

$$\frac{dx}{dy} = \frac{2y}{4a} = \frac{y}{2a}$$

$$\left(\frac{dx}{dy}\right)^2 = \frac{y^2}{4a^2}$$

$$1 + \left(\frac{dx}{dy}\right)^2 = 1 + \frac{y^2}{4a^2}$$

$$1 + \left(\frac{dx}{dy}\right)^2 = \frac{4a^2 + y^2}{4a^2}$$

$$\sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \frac{\sqrt{4a^2 + y^2}}{2a}$$

$$\frac{a^2 \ln(y + \sqrt{a^2 + y^2})}{a}$$

7.6-5

$$= \frac{1}{2a} \left(\frac{3a}{4} \left(\frac{5a}{2} \right) + 2a^2 \ln \left(\frac{\frac{3a}{2} + \frac{5a}{2}}{2a} \right) \right)$$

$$= \frac{1}{2a} \left(\frac{15a^2}{8} + 2a^2 \ln \left(\frac{8a}{2a} \right) \right)$$

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$$= \frac{15a^2}{16a} + \frac{2a^2}{2a} \ln \left(\frac{8a}{4a} \right)$$

$$= \frac{15a}{16} + a \ln(2a) = a \left(\frac{15}{16} + \ln 2 \right) \text{ Ans.}$$

(2)

$$y = c \cosh \frac{x}{c}$$

$$S' = \int_a^b \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$$

$$= \int_0^x \sqrt{1 + \sinh^2 \frac{x}{c}} dx$$

$$= \int_0^x \cosh \frac{x}{c} dx$$

$$= \int_0^x \cosh \frac{x}{c} dx$$

$$= \left| \frac{\sinh \frac{x}{c}}{\frac{1}{c}} \right|_0^x$$

$$y = c \cosh \frac{x}{c}$$

$$\frac{dy}{dx} = c \sinh \frac{x}{c} \cdot \frac{1}{c}$$

$$\left(\frac{dy}{dx} \right)^2 = \sinh^2 \frac{x}{c}$$

$$1 + \left(\frac{dy}{dx} \right)^2 = 1 + \sinh^2 \frac{x}{c}$$

$$\text{we know } \cosh^2 \frac{x}{c} - \sinh^2 \frac{x}{c} = 1$$

$$S' = \left[c \sinh \frac{x}{c} - 0 \right] = c \sinh \frac{x}{c} \text{ Ans.}$$

$$= 12 (e^2 - 1) \text{ Ans.}$$

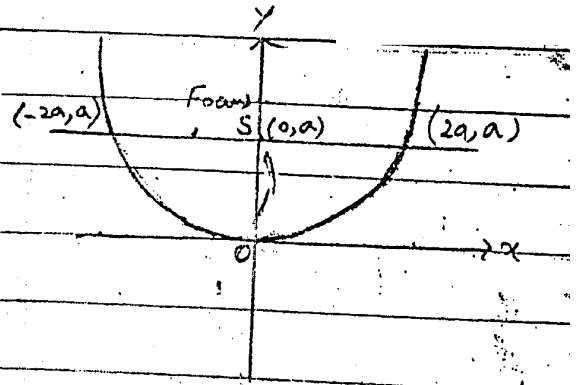


7.6-6

Q3 $x^2 = 4ay$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$



$$S = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \int_0^{2a} \sqrt{1 + \frac{x^2}{4a^2}} dx = \int_0^{2a} \sqrt{\frac{4a^2 + x^2}{4a^2}} dx$$

$$= \frac{1}{2a} \int_0^{2a} \sqrt{(2a)^2 + x^2} dx = \frac{1}{2a} \left[\frac{\sqrt{(2a)^2 + x^2} \cdot x}{2} + \frac{(2a)^2}{2} \ln \left| \frac{x + \sqrt{(2a)^2 + x^2}}{2a} \right| \right]_0^{2a}$$

$$= \frac{1}{2a} \left[\frac{\sqrt{(2a)^2 + (2a)^2} \cdot 2a}{2} + \frac{(2a)^2}{2} \ln \left| \frac{2a + \sqrt{(2a)^2 + (2a)^2}}{2a} \right| \right] - \frac{1}{2a} \left[0 + 2a^2 \ln \left| \frac{0 + \sqrt{4a^2}}{2a} \right| \right]$$

$$= \frac{1}{2a} \left[\sqrt{8} a + 2a^2 \ln \left| \frac{2a + \sqrt{8a^2}}{2a} \right| \right] - \frac{1}{2a} \left[0 + 2a^2 \ln 1 \right]$$

$$= \frac{1}{2a} \left[2a\sqrt{2} a + 2a^2 \ln \left| \frac{2a(1 + \sqrt{2})}{2a} \right| \right] - \frac{1}{2a} \left[0 + 2a^2 (0) \right]$$

$$= \frac{1}{2a} \left[2a^2\sqrt{2} + 2a^2 \ln (1 + \sqrt{2}) \right] - 0$$

$$= \frac{2a^2}{2a} \left[\sqrt{2} + \ln (1 + \sqrt{2}) \right]$$

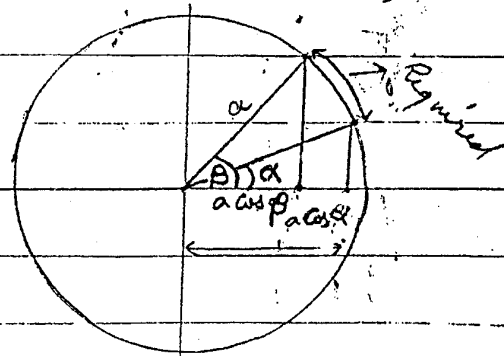
$$= a \left[\sqrt{2} + \ln (1 + \sqrt{2}) \right]$$

7.5-8

Q.5. $x^2 + y^2 = a^2$

$x = a \cos \alpha, \quad y = a \sin \beta$

$$S = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$



Parametric Eqs of circle are

$x = a \cos \theta, \quad y = a \sin \theta$

$\frac{dx}{d\theta} = -a \sin \theta$

$\frac{dy}{d\theta} = a \cos \theta$

$\left(\frac{dx}{d\theta}\right)^2 = a^2 \sin^2 \theta$

$\left(\frac{dy}{d\theta}\right)^2 = a^2 \cos^2 \theta$

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$$S = \int_{\alpha}^{\beta} \sqrt{a^2 \sin^2 \theta + a^2 \cos^2 \theta} d\theta = \int_{\alpha}^{\beta} \sqrt{a^2 (\sin^2 \theta + \cos^2 \theta)} d\theta$$

$$= \int_{\alpha}^{\beta} a d\theta = a \left| \theta \right|_{\alpha}^{\beta} = a (\beta - \alpha) \text{ Ans.}$$

2nd Method

$x^2 + y^2 = a^2$

$y = \sqrt{a^2 - x^2}$

$$S = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$\frac{dy}{dx} = \frac{1}{2} \frac{(-2x)}{\sqrt{a^2 - x^2}}$

$= \int_{a \cos \beta}^{a \cos \alpha} \sqrt{\frac{a^2}{a^2 - x^2}} dx$

$= a \int_{a \cos \beta}^{a \cos \alpha} \frac{dx}{\sqrt{a^2 - x^2}}$

$\left(\frac{dy}{dx}\right)^2 = \frac{x^2}{a^2 - x^2}$

$= a \left[-\cos^{-1} \frac{x}{a} \right]_{a \cos \beta}^{a \cos \alpha}$

$= a \left[-\cos^{-1} \frac{a \cos \alpha}{a} + \cos^{-1} \frac{a \cos \beta}{a} \right]$

$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{x^2}{a^2 - x^2} = \frac{a^2 - x^2 + x^2}{a^2 - x^2} = \frac{a^2}{a^2 - x^2}$

$= a \left[-\cos^{-1} \cos \alpha + \cos^{-1} \cos \beta \right] = a (-\alpha + \beta) \text{ Ans.}$

7.6 - 10

$$= \int_0^{2\pi} \sqrt{a^2(2-2\cos\theta)} d\theta$$

$$= \int_0^{2\pi} \sqrt{a^2(1-\cos\theta)} d\theta = \int_0^{2\pi} \sqrt{2a^2 \sin^2 \frac{\theta}{2}} d\theta$$

$$= \int_0^{2\pi} 2a \sin \frac{\theta}{2} d\theta = 2a \left[-\frac{\cos \frac{\theta}{2}}{1/2} \right]_0^{2\pi}$$

$$= 4a \left[\cos \frac{2\pi}{2} + \cos 0 \right]$$

$$= 4a (\cos \pi + 1) = 4a (-(-1) + 1) = 8a$$

Q8. $x^{2/3} + y^{2/3} = a^{2/3}$

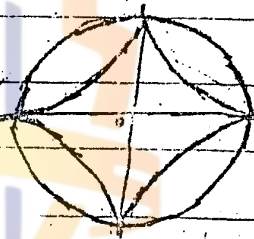
Parametric Eqs of Astroid

$$x = a \cos^3 \theta$$

$$y = a \sin^3 \theta$$

$$\frac{dx}{d\theta} = 3a \cos^2 \theta (-\sin \theta)$$

$$\frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$



$0 \rightarrow \frac{\pi}{2}$
one quad

$$S = \int_0^{\frac{\pi}{2}} \sqrt{9a^2 \cos^4 \theta \sin^2 \theta + 9a^2 \sin^4 \theta \cos^2 \theta} d\theta$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{9a^2 \cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta)} d\theta$$

$$= \int_0^{\frac{\pi}{2}} 3a \cos \theta \sin \theta d\theta = 3a \left[\frac{\sin^2 \theta}{2} \right]_0^{\frac{\pi}{2}}$$

$$= 3a \left[\frac{\sin^2 \frac{\pi}{2}}{2} \right] = 3a \left[\frac{1}{2} \right] = \frac{3a}{2} \text{ Proved.}$$

7.6-11

(ii) $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$

$x = a \cos^3 \theta \quad y = b \sin^3 \theta$

$\frac{dx}{d\theta} = 3a \cos^2 \theta (-\sin \theta)$

$\frac{dy}{d\theta} = 3b \sin^2 \theta \cos \theta$

$\left(\frac{dx}{d\theta}\right)^2 = 9a^2 \cos^4 \theta \sin^2 \theta$

$\left(\frac{dy}{d\theta}\right)^2 = 9b^2 \sin^4 \theta \cos^2 \theta$

$S = \int_0^{\pi/2} \sqrt{9a^2 \cos^4 \theta \sin^2 \theta + 9b^2 \sin^4 \theta \cos^2 \theta} d\theta$

$= \int_0^{\pi/2} \sqrt{9 \cos^2 \theta \sin^2 \theta (a^2 \cos^2 \theta + b^2 \sin^2 \theta)} d\theta$

$= \int_0^{\pi/2} 3 \cos \theta \sin \theta \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} d\theta$

Put $a^2 \cos^2 \theta + b^2 \sin^2 \theta = t^2$

when $\theta = 0, t = a$

$\theta = \pi/2, t = b$

$\{2a^2 \cos \theta (-\sin \theta) + 2b^2 \sin \theta \cos \theta\} d\theta = 2t dt$

$2 \sin \theta \cos \theta (b^2 - a^2) d\theta = 2t dt$

$\sin \theta \cos \theta d\theta = \frac{2t dt}{2(b^2 - a^2)}$

$\int_a^b \frac{t^2}{b^2 - a^2} 3t dt = \int_a^b \frac{3t^3 dt}{b^2 - a^2} = \left| \frac{3t^4}{4(b^2 - a^2)} \right|_a^b$

$= \frac{b^4}{b^2 - a^2} - \frac{a^4}{b^2 - a^2} = \frac{b^4 - a^4}{b^2 - a^2} = \frac{(b^2 - a^2)(b^2 + a^2)}{b^2 - a^2}$

$= \frac{b^2 + a^2}{1} \text{ Ans.}$

7.6-12

Q9. $r = a(1 - \cos \theta)$

$\frac{dr}{d\theta} = a(\sin \theta)$

$S = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$

$= 2 \int_0^{\pi} \sqrt{a^2(1 - \cos \theta)^2 + a^2 \sin^2 \theta} d\theta$

$= 2 \int_0^{\pi} \sqrt{a^2(1 + \cos^2 \theta - 2\cos \theta + \sin^2 \theta)} d\theta$

$= 2 \int_0^{\pi} \sqrt{a^2(1 + 1 - 2\cos \theta)} d\theta$

$= 2 \int_0^{\pi} \sqrt{a^2 2(1 - \cos \theta)} d\theta = 2 \int_0^{\pi} \sqrt{2a^2 2 \sin^2 \frac{\theta}{2}} d\theta$

$= 2 \int_0^{\pi} 2a \sin \frac{\theta}{2} d\theta = 4a \left[-\cos \frac{\theta}{2} \right]_0^{\pi} = 8a \left[-\cos \frac{\pi}{2} + \cos 0 \right]$

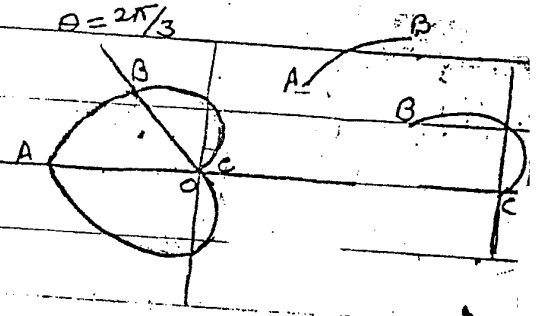
$= 8a(0 + 1) = 8a$ = Entire length of Cardioid

Now length of the arc from $\theta = 0$ to $\theta = \frac{2\pi}{3}$

$S_2 = \int_0^{\frac{2\pi}{3}} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_0^{\frac{2\pi}{3}} 2a \sin \frac{\theta}{2} d\theta$ solved above

$= 2a \left[-\cos \frac{\theta}{2} \right]_0^{\frac{2\pi}{3}} = 4a \left[\cos \frac{2\pi}{3} + \cos 0 \right]$

$= 4a \left[-\frac{1}{2} + 1 \right] = 4a \left(\frac{1}{2} \right) = 2a$ Ans.



$0 \rightarrow \pi$ for upper half only

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7.6 - 13

11. $r = a(1 + \cos \theta)$ — (1)

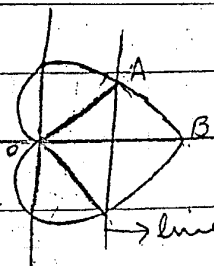
$4a \cos \theta = 3a$ — (11)

To find the upper & lower limits

from (11) $r = \frac{3a}{4 \cos \theta}$ Put in (1)

$\frac{3a}{4 \cos \theta} = a(1 + \cos \theta)$

The curve is symmetrical about initial line. θ by $(-\theta)$ No change.



3a = 4a cos theta (1 + cos theta)

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3a = 4a cos theta + 4a cos^2 theta

$4 \cos^2 \theta + 4 \cos \theta - 3 = 0$

Put $\cos \theta = t$

$4t^2 + 4t - 3 = 0$

$4t^2 + 6t - 2t - 3 = 0$

$2t(2t+3) - 1(2t+3) = 0$

$(2t+3)(2t-1) = 0$

$t = -\frac{3}{2}, t = \frac{1}{2}$

$\cos \theta = -\frac{3}{2}, \frac{1}{2}$

$\cos \theta$ can never be $-\frac{3}{2}$ $\therefore$ limit of $\cos \theta$

is from -1 to 1 .

$\therefore \cos \theta = \frac{1}{2}, \therefore \theta = \frac{\pi}{3}, -\frac{\pi}{3}$

$\cos \theta = -\frac{3}{2}, \frac{1}{2}$

11

Length of arc from B to A is $S = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$

$S = \int_0^{\frac{\pi}{3}} \sqrt{a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta} d\theta$

$r = a(1 + \cos \theta)$

$\frac{dr}{d\theta} = a(-\sin \theta)$

$= \int_0^{\frac{\pi}{3}} \sqrt{a^2(1 + \cos^2 \theta + 2 \cos \theta + \sin^2 \theta)} d\theta$

$\left(\frac{dr}{d\theta}\right)^2 = a^2 \sin^2 \theta$

$= \int_0^{\frac{\pi}{3}} \sqrt{a^2(1 + 1 + 2 \cos \theta)} d\theta$

$r^2 + \left(\frac{dr}{d\theta}\right)^2 = a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta$

7.6-14

$$= \int_0^{\pi/3} \sqrt{2a^2(1+\cos\theta)} d\theta = \int_0^{\pi/3} \sqrt{2a^2 \cdot 2\cos^2\frac{\theta}{2}} d\theta$$

$$= \int_0^{\pi/3} 2a \cos\frac{\theta}{2} d\theta = 2a \left/ \frac{\sin\theta}{\frac{1}{2}} \right/ \Big|_0^{\pi/3}$$

$$S_1 = 4a \left[\sin\frac{\pi}{6} - \sin 0 \right] = 4a \left(\frac{1}{2} - 0 \right) = 2a.$$

Now length of arc from A to O

$$S_2 = \int_{\pi/3}^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_{\pi/3}^{\pi} \sqrt{a^2(1+\cos\theta)^2 + a^2 \sin^2\theta} d\theta$$

$$= \int_{\pi/3}^{\pi} 2a \cos\frac{\theta}{2} d\theta. \quad \text{as solved above}$$

$$= 2a \left/ \frac{\sin\theta}{\frac{1}{2}} \right/ \Big|_{\pi/3}^{\pi} = 4 \left[\sin\frac{\pi}{2} - \sin\frac{\pi}{6} \right]$$

$$S_2 = 4a \left(1 - \frac{1}{2} \right) = 2a.$$

$$S_1 = S_2 \quad \text{Ratio } 2a : 2a = 1:1$$

This shows that line $4r\cos\theta = 3a$ bisect upper half of the curve at $\theta = \pi/3$. Since curve is symmetric therefore similarly line bisect lower half at $\theta = -\pi/3$.

$\therefore 4r\cos\theta = 3a$ bisects the cardioid in ratio 1:1.

7-6-15

Q.12 $x = (a+b)\cos\theta - b\cos\left(\frac{a+b}{b}\theta\right)$

$y = (a+b)\sin\theta - b\sin\left(\frac{a+b}{b}\theta\right)$

$\frac{dx}{d\theta} = (a+b)(-\sin\theta) - b(-\sin\left(\frac{a+b}{b}\theta\right))\left(\frac{a+b}{b}\right)$

$= (a+b)\left[\sin\left(\frac{a+b}{b}\theta\right) - \sin\theta\right]$

$\left(\frac{dx}{d\theta}\right)^2 = (a+b)^2\left[\sin\left(\frac{a+b}{b}\theta\right) - \sin\theta\right]^2$

$\left(\frac{dy}{d\theta}\right) = (a+b)\cos\theta - b\cos\left(\frac{a+b}{b}\theta\right)\left(\frac{a+b}{b}\right)$

$= (a+b)\left[\cos\theta - \cos\left(\frac{a+b}{b}\theta\right)\right]$

$\left(\frac{dy}{d\theta}\right)^2 = (a+b)^2\left[\cos\theta - \cos\left(\frac{a+b}{b}\theta\right)\right]^2$

$S = \int_{\frac{\pi b}{a}}^{\theta} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$

$= \int_{\frac{\pi b}{a}}^{\theta} \sqrt{(a+b)^2\left[\sin\left(\frac{a+b}{b}\theta\right) - \sin\theta\right]^2 + (a+b)^2\left[\cos\theta - \cos\left(\frac{a+b}{b}\theta\right)\right]^2} d\theta$

$= \int_{\frac{\pi b}{a}}^{\theta} (a+b)^2 \left[\sin^2\left(\frac{a+b}{b}\theta\right) + \sin^2\theta - 2\sin\left(\frac{a+b}{b}\theta\right)\sin\theta + \cos^2\theta + \cos^2\left(\frac{a+b}{b}\theta\right) - 2\cos\theta\cos\left(\frac{a+b}{b}\theta\right) \right] d\theta$

$= \int_{\frac{\pi b}{a}}^{\theta} (a+b)^2 \left[\left(\sin^2\left(\frac{a+b}{b}\theta\right) + \cos^2\left(\frac{a+b}{b}\theta\right)\right) + \left(\sin^2\theta + \cos^2\theta\right) - 2\sin\left(\frac{a+b}{b}\theta\right)\sin\theta - 2\cos\theta\cos\left(\frac{a+b}{b}\theta\right) \right] d\theta$

7.6-16

$$= (a+b) \int_{\frac{\pi b}{a}}^{\theta} \sqrt{1 - 2 \left(\sin\left(\frac{a+b}{b}\theta\right) \sin \theta + \cos\left(\frac{a+b}{b}\theta\right) \cos \theta \right)} d\theta.$$

$$= (a+b) \int_{\frac{\pi b}{a}}^{\theta} \sqrt{2 - 2 \left(\cos\left\{\frac{a+b}{b}\theta - \theta\right\}\right)} d\theta.$$

$$= (a+b) \int_{\frac{\pi b}{a}}^{\theta} \sqrt{2 \left(1 - \cos\left\{\frac{a+b}{b}\theta - \theta\right\}\right)} d\theta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= (a+b) \int_{\frac{\pi b}{a}}^{\theta} \sqrt{2 \left(1 - \cos\left(\frac{a\theta}{b}\right)\right)} d\theta.$$

$$1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$$

$$= (a+b) \int_{\frac{\pi b}{a}}^{\theta} \sqrt{2 \left(2 \sin^2 \left(\frac{a\theta}{2b} \right) \right)} d\theta = (a+b) 2 \int_{\frac{\pi b}{a}}^{\theta} \frac{\sin a\theta}{2b} d\theta.$$

$$= 2(a+b) \left[-\frac{\cos \frac{a\theta}{2b}}{\frac{a}{2b}} \right]_{\frac{\pi b}{a}}^{\theta} = 2(a+b) \frac{2b}{a} \left[-\cos\left(\frac{a\theta}{2b}\right) \right]_{\frac{\pi b}{a}}^{\theta}$$

$$= \frac{4b}{a} (a+b) \left(-\cos \frac{a\theta}{2b} + \cos \frac{a \frac{\pi b}{a}}{2b} \right)$$

$$= \frac{4b(a+b)}{a} \left(-\cos \frac{a\theta}{2b} + \cos \frac{\pi}{2} \right)$$

$$\cos \frac{\pi}{2} = 0$$

$$= -\frac{4b(a+b)}{a} \cos \frac{a\theta}{2b}$$

$$= \frac{4b(a+b)}{a} \cos\left(\frac{a\theta}{2b}\right)$$

Ignore the sign

7.6-17

one loop if $a > b$

$$\odot r = 3 + 2\cos\theta \quad \odot r = 3 - 2\cos\theta$$

$$r = a \pm b \sin\theta$$

Two loops if $a < b$

$$\odot r = 2 + 3\sin\theta$$

$$\odot r = 2 - 3\sin\theta$$

one loop if $a > b$

$$\odot r = 3 + 2\sin\theta$$

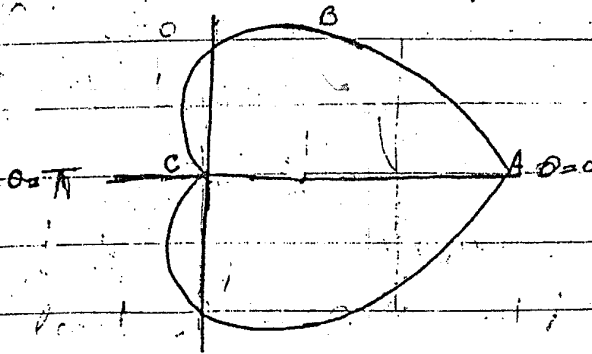
$$\odot r = 3 - 2\sin\theta$$

Q13

$$r = a + b \cos\theta$$

$\frac{b}{a}$ is small $\therefore b < a$

It is symmetric about initial line $\therefore \theta = -\theta$ No change.



$$r = 0 \quad 0 = a + b \cos\theta$$

$$-a = b \cos\theta$$

$$-\frac{a}{b} = \cos\theta$$

$$\text{Minimum value for } \cos\theta = -1 \Rightarrow \theta = \pi$$

$$\text{Total length of perimeter} = 2 \int_0^\pi \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \quad \therefore 2ABC$$

$$r = a + b \cos\theta$$

$$\frac{dr}{d\theta} = -b \sin\theta$$

$$r^2 + \left(\frac{dr}{d\theta}\right)^2 = (a + b \cos\theta)^2 + b^2 \sin^2\theta$$

$$= a^2 + b^2 \cos^2\theta + 2ab \cos\theta + b^2 \sin^2\theta$$

$$= a^2 + 2ab \cos\theta + b^2 (\cos^2\theta + \sin^2\theta)$$

$$= a^2 + 2ab \cos\theta + b^2$$

$$\therefore \frac{b}{a} = K$$

$$= a^2 \left[1 + 2 \frac{b \cos\theta}{a} + \frac{b^2}{a^2} \right] = a^2 (1 + 2K \cos\theta + K^2)$$

$$2 \int_0^\pi \sqrt{a^2 (1 + 2K \cos\theta + K^2)} d\theta = 2a \int_0^\pi (1 + 2K \cos\theta + K^2)^{\frac{1}{2}} d\theta$$

7.6-18

Apply Binomial Series $(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \dots$

$$= 2a \int_0^\pi \left(1 + \frac{1}{2}(2K \cos \theta + K^2) + \frac{1}{2} \left(\frac{1}{2} - 1 \right) (2K \cos \theta + K^2)^2 + \dots \right) d\theta$$

neglecting $K^3, K^4, \dots$
K is small

$$= 2a \int_0^\pi \left(1 + K \cos \theta + \frac{K^2}{2} - \frac{1}{8}(4K^2 \cos^2 \theta + K^4 + 4K^3 \cos \theta) \right) d\theta$$

$$= 2a \int_0^\pi \left(1 + K \cos \theta + \frac{K^2}{2} - \frac{K^2}{2} \cos^2 \theta \right) d\theta$$

$$= 2a \int_0^\pi \left(1 + K \cos \theta + \frac{K^2}{2} (1 - \cos^2 \theta) \right) d\theta$$

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$$= 2a \int_0^\pi \left(1 + K \cos \theta + \frac{K^2}{2} \sin^2 \theta \right) d\theta$$

$$= 2a \int_0^\pi d\theta + 2a \left(K \cos \theta + \frac{2aK^2}{\pi} \right) \int_0^\pi \sin^2 \theta d\theta$$

$$= 2a \left| \theta \right|_0^\pi + 2aK \left| \sin \theta \right|_0^\pi + \frac{aK^2}{2} \left| \theta - \frac{\sin 2\theta}{2} \right|_0^\pi$$

$$= 2a(\pi - 0) + 2aK(\sin \pi - \sin 0) + \frac{aK^2}{2} \left(\pi - \frac{\sin 2\pi}{2} - 0 + 0 \right)$$

$$= 2a\pi + 2aK(0) + \frac{aK^2}{2} \pi$$

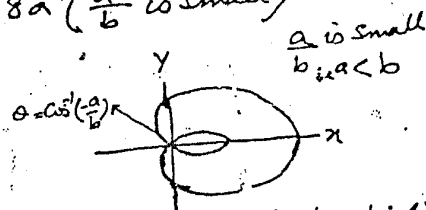
$$= 2a\pi + \frac{aK^2 \pi}{2} = 2a\pi \left(1 + \frac{aK^2 \pi}{2 \cdot 2a\pi} \right)$$

$$= 2a\pi \left(1 + \frac{K^2}{4} \right) = 2a\pi \left(1 + \frac{b^2}{4a^2} \right) \text{ Proved}$$

7.6-19

Q14 Prove that the difference between the lengths of the two loops of the limaçon $r = a + b \cos \theta$ is $8a$ ($\frac{a}{b}$ is small)

Sol $r = a + b \cos \theta$ — (i)
 Put $r = 0 \Rightarrow \cos \theta = -\frac{a}{b}$
 $\theta = \cos^{-1}(-\frac{a}{b})$



① is symmetric about initial line
 The upper half of outer loop varies from 0 to $\cos^{-1}(-\frac{a}{b})$
 The lower half of inner loop varies from $\cos^{-1}(-\frac{a}{b})$ to π .

$$S = 2 \left[\int_0^{\cos^{-1}(-\frac{a}{b})} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta - \int_{\cos^{-1}(-\frac{a}{b})}^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \right] \text{ — (ii)}$$

$$\int_0^{\cos^{-1}(-\frac{a}{b})} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_0^{\cos^{-1}(-\frac{a}{b})} \sqrt{(a + b \cos \theta)^2 + b^2 \sin^2 \theta} d\theta$$

$$= \int_0^{\cos^{-1}(-\frac{a}{b})} \sqrt{a^2 + b^2 (\cos^2 \theta + \sin^2 \theta) + 2ab \cos \theta} d\theta$$

$$= \int_0^{\cos^{-1}(-\frac{a}{b})} \sqrt{a^2 + b^2 + 2ab \cos \theta} d\theta$$

$$= \int_0^{\cos^{-1}(-\frac{a}{b})} b \left(1 + \frac{2a}{b} \cos \theta \right)^{\frac{1}{2}} d\theta$$

$$= b \left[\left(1 + \frac{2a}{b} \cos \theta \right)^{\frac{1}{2}} \right]_0^{\cos^{-1}(-\frac{a}{b})}$$

$$= b \left[\left(1 + \frac{2a}{b} \cos \theta \right)^{\frac{1}{2}} \right]_0^{\cos^{-1}(-\frac{a}{b})}$$

$$= b \left[\theta + \frac{a}{b} \sin \theta \right]_0^{\cos^{-1}(-\frac{a}{b})}$$

$$= b \left[\theta + \frac{a}{b} \sin \theta \right] \text{ — (iii)}$$

Similarly $\int_{\cos^{-1}(-\frac{a}{b})}^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = b \left[\theta + \frac{a}{b} \sin \theta \right]_{\cos^{-1}(-\frac{a}{b})}^{\pi}$

$$= b \left[\pi + 0 - \theta - \frac{a}{b} \sin \theta \right] \text{ — (iv) Put (iii) + (iv) in (ii)}$$

$$S = 2b \left[\theta + \frac{a}{b} \sin \theta - \pi + \theta + \frac{a}{b} \sin \theta \right]$$

$$= 2b \left[2\theta - \pi + 2 \frac{a}{b} \sin \theta \right]$$

$$= 2b \left[2 \cos^{-1}(-\frac{a}{b}) - \pi + 2 \frac{a}{b} \left(\sqrt{1 - \cos^2 \theta} \right) \right]$$

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \\ \sin^2 \theta &= 1 - \cos^2 \theta \\ \sin \theta &= \sqrt{1 - \cos^2 \theta} \end{aligned}$$

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7.6-20

$$S = 2b \left[2 \left(\frac{\pi}{2} + \frac{a}{b} \right) - \pi + 2 \frac{a}{b} \cdot \sqrt{1 - \frac{a^2}{b^2}} \right]$$

$$= 2b \left[2 \left(\frac{\pi}{2} + \frac{a}{b} \right) - \pi + \frac{2a}{b} \sqrt{1 - \frac{a^2}{b^2}} \right] \quad \text{Neglecting } \frac{a^2}{b^2} \text{ as } \frac{a}{b} \text{ is small.}$$

$$= 2b \left[\pi + \frac{2a}{b} - \pi + \frac{2a}{b} \right]$$

$$= 2b \left[\frac{4a}{b} \right]$$

$$= 8a \text{ Ans}$$

$$\because \theta = \cos^{-1} \left(-\frac{a}{b} \right) = \frac{\pi}{2} + \frac{a}{b}$$

$$\cos^{-1} \left(-\frac{a}{b} \right) = \frac{\pi}{2} + \frac{a}{b}$$

$$\text{Proof } \cos \left(\frac{\pi}{2} + \frac{a}{b} \right)$$

$$= \cos \frac{\pi}{2} \cos \frac{a}{b} - \sin \frac{\pi}{2} \sin \frac{a}{b}$$

$$\cos \left(\frac{\pi}{2} + \frac{a}{b} \right) = 0 - \sin \frac{a}{b}$$

$$\cos \left(\frac{\pi}{2} + \frac{a}{b} \right) = -\frac{a}{b} \quad \left(\because \sin \theta = \theta \text{ when } \theta \text{ is small} \right)$$

$$\frac{\pi}{2} + \frac{a}{b} = \cos^{-1} \left(-\frac{a}{b} \right)$$

x

Q3 Find length of loop of curve $3ay^2 = x(a-x)^2$

Sol Put y by $-y$, curve is unchanged so it is symmetric about x -axis.

$$\text{Put } y = 0 \Rightarrow 0 = x(a-x)^2 \Rightarrow x = 0, a$$

$$S = 2 \int_0^a \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$$

$$= 2 \int_0^a \sqrt{1 + \frac{(a-3x)^2}{12ax}} dx$$

$$= 2 \int_0^a \frac{\sqrt{12ax + a^2 + 9x^2 - 6ax}}{12ax} dx$$

$$= 2 \int_0^a \frac{\sqrt{(3x+a)^2}}{12ax} dx$$

$$= \frac{2}{12a} \int_0^a \frac{(3x+a)}{\sqrt{x}} dx$$

$$= \frac{1}{6a} \int_0^a (3\sqrt{x} + a x^{-\frac{1}{2}}) dx$$

$$= \frac{1}{6a} \left[3x^{\frac{3}{2}} + a x^{\frac{1}{2}} \right]_0^a$$

$$= \frac{1}{6a} \left(2a^{\frac{3}{2}} + 2a a^{\frac{1}{2}} - 0 - 0 \right)$$

$$= \frac{1}{6a} (4a^{\frac{3}{2}}) = \frac{4}{6} a^{\frac{3}{2}-1} = \frac{4a}{3} \text{ Ans}$$

$$3ay^2 = x(a-x)^2$$

$$\sqrt{3a} y = \sqrt{x} (a-x)$$

$$\sqrt{3a} y = \sqrt{x} a - x^{\frac{3}{2}}$$

$$\sqrt{3a} \frac{dy}{dx} = \frac{a}{2\sqrt{x}} - \frac{3\sqrt{x}}{2}$$

$$\sqrt{3a} \frac{dy}{dx} = \frac{a-3x}{2\sqrt{x}}$$

$$\frac{dy}{dx} = \frac{(a-3x)}{\sqrt{3a} \cdot 2\sqrt{x}}$$

$$\left(\frac{dy}{dx} \right)^2 = \frac{(a-3x)^2}{12ax}$$

7.6-21

Q10 $r = \sin^2 \frac{\theta}{2}$ from $(0,0)$ to $(1,\pi)$

$$\frac{dr}{d\theta} = \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cdot \frac{1}{2}$$

$$\left(\frac{dr}{d\theta}\right)^2 = \sin^2 \frac{\theta}{2} \cos^2 \frac{\theta}{2}$$

$$\int_0^\pi \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_0^\pi \sqrt{\left(\sin^2 \frac{\theta}{2}\right)^2 + \sin^2 \frac{\theta}{2} \cos^2 \frac{\theta}{2}} d\theta$$

$$= \int_0^\pi \sqrt{\sin^4 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} (1 - \sin^2 \frac{\theta}{2})} d\theta = \int_0^\pi \sqrt{\sin^4 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} - \sin^4 \frac{\theta}{2}} d\theta$$

$$= \int_0^\pi \sin \frac{\theta}{2} d\theta = \left| -\frac{\cos \frac{\theta}{2}}{1/2} \right|_0^\pi = \left| -2 \cos \frac{\theta}{2} \right|_0^\pi$$

$$= \left[-2 \cos \frac{\pi}{2} + 2 \cos \frac{0}{2} \right] = (0 + 2) = 2 \text{ Ans.}$$

Q15 $a = r\theta$ $r = \frac{a}{\theta}$ $\frac{dr}{d\theta} = -\frac{a}{\theta^2}$

$$S = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

when $r = a$ $\theta = \frac{a}{a} = 1$

when $r = 2a$ $\theta = \frac{a}{2a} = \frac{1}{2}$

$$S = \int_{1/2}^1 \sqrt{\frac{a^2}{\theta^2} + \frac{a^2}{\theta^4}} d\theta$$

$$= \int_{1/2}^1 \sqrt{\frac{a^2 \theta^2 + a^2}{\theta^4}} d\theta = \int_{1/2}^1 \frac{a^2 (\theta^2 + 1)}{\theta^4} d\theta$$

$$= a \int_{1/2}^1 \frac{\sqrt{\theta^2 + 1}}{\theta^2} \cdot \frac{\theta^2 + 1}{\sqrt{\theta^2 + 1}} d\theta = a \int_{1/2}^1 \frac{\theta^2 + 1}{\theta^2 \sqrt{\theta^2 + 1}} d\theta$$

IInd Method

$$S = \int_a^{2a} \sqrt{1 + \left(\frac{r}{dr}\right)^2} dr$$

$$= \int_a^{2a} \frac{\sqrt{r^2 + a^2}}{r} dr \quad \text{Put } r = a \tan \alpha$$

$$= a \int_{\pi/4}^{\pi/2} \sec^2 \alpha \cos \alpha d\alpha \quad \text{IB}$$

$$\times \frac{15-1}{2(\sqrt{2}-1)} = \frac{15-1}{2(\sqrt{2}-1)} \cdot \frac{15+1}{15+1} \cdot \frac{\sqrt{2}+1}{\sqrt{2}+1} \quad \text{Rationalizing}$$

7.6-22

$$= \frac{1}{2} \frac{(5-1)(\sqrt{2}+1)}{15+1(2-1)} = \frac{4}{2} \frac{(\sqrt{2}+1)}{15+1} = \frac{2(\sqrt{2}+1)}{15+1} = \frac{2\sqrt{2}+2}{15+1} = \frac{\sqrt{2}+1}{5+1}$$

$$= a \int \frac{\theta^2 d\theta}{\theta^2 \sqrt{\theta^2+1}} + a \int \frac{d\theta}{\theta^2 \sqrt{\theta^2+1}}$$

$$\int \frac{\sqrt{a^2+x^2}}{x} dx = a \sqrt{1+\cot^2 x} \tan x + a \ln \left| \frac{1+\cot^2 x - \cot x}{1+\cot^2 x + \cot x} \right|$$

$$\int \frac{\sqrt{a^2+x^2}}{x} dx = \left| a \sqrt{1+\left(\frac{a}{x}\right)^2} \frac{x}{a} + a \ln \left| \frac{1+\left(\frac{a}{x}\right)^2 - \frac{a}{x}}{1+\left(\frac{a}{x}\right)^2 + \frac{a}{x}} \right| \right|_a^{2a}$$

$$= a \int \frac{d\theta}{\sqrt{\theta^2+1}} + a \int \frac{d\theta}{\theta^2 \sqrt{\theta^2+1}}$$

$$\text{Now } a \int \frac{d\theta}{\sqrt{\theta^2+1}} = a \ln(\theta + \sqrt{\theta^2+1})$$

$$\begin{aligned} &= \left| a \sqrt{1+\left(\frac{a}{x}\right)^2} \frac{x}{a} + a \ln \left| \frac{1+\left(\frac{a}{x}\right)^2 - \frac{a}{x}}{1+\left(\frac{a}{x}\right)^2 + \frac{a}{x}} \right| \right|_a^{2a} \\ &= \left[a\sqrt{5} + a \ln \left| \frac{a\sqrt{5}-a}{2a} \right| \right] - \left[a\sqrt{2} + a \ln \left| \frac{a\sqrt{2}-a}{a} \right| \right] \\ &= a(\sqrt{5}-\sqrt{2}) + a \ln(\sqrt{5}-1) - a \ln(\sqrt{2}-1) \\ &= a\left[\sqrt{5}-\sqrt{2} + \ln \frac{\sqrt{5}-1}{\sqrt{2}-1} \right] = a\left[\sqrt{5}-\sqrt{2} + \ln \frac{2+\sqrt{5}}{1+\sqrt{2}} \right] \end{aligned}$$

$$= a \left[\ln \left(\frac{1}{2} + \sqrt{\frac{1}{4}+1} \right) - \ln(1+\sqrt{1}) \right] = a \left[\ln \left(\frac{1}{2} + \sqrt{\frac{5}{4}} \right) - \ln(1+\sqrt{2}) \right] \quad \text{See above}$$

$$= a \left[\ln \left(\frac{1+\sqrt{5}}{2} \right) - \ln(1+\sqrt{2}) \right] = a \ln \left(\frac{1+\sqrt{5}}{2(1+\sqrt{2})} \right)$$

$$\text{Consider } a \int \frac{d\theta}{\theta^2 \sqrt{\theta^2+1}}$$

$$\text{Put } \theta = \frac{1}{t} \quad d\theta = -\frac{1}{t^2} dt \quad \text{when } \theta=1, t=1$$

$$\theta=\frac{1}{2}, t=2$$

$$= a \int \frac{(-\frac{1}{t^2}) dt}{\left(\frac{1}{t^2}\right) \sqrt{\frac{1}{t^2}+1}} = a \int \frac{-dt}{\sqrt{1+t^2}} = a \int \frac{-t dt}{1+t^2} = -\frac{a}{2} \int \frac{(1+t^2)^{\frac{1}{2}}}{t^2} dt$$

$$= -\frac{a}{2} \left[\frac{(1+t^2)^{\frac{1}{2}}}{\frac{1}{t}} \right]_1^2 = -\frac{a}{2} \left[\frac{(1+4)^{\frac{1}{2}}}{2} - \frac{(1+1)^{\frac{1}{2}}}{1} \right] = -\frac{a}{2} [\sqrt{5}-\sqrt{2}]$$

$$\therefore \text{It becomes } S = a \ln \left(\frac{1+\sqrt{5}}{2(1+\sqrt{2})} \right) - a[\sqrt{5}-\sqrt{2}]$$

$$= a \ln \left(\frac{2(1+\sqrt{2})}{1+\sqrt{5}} \right) - a(\sqrt{5}-\sqrt{2}) = -a \ln \left(\frac{2(1+\sqrt{2})}{1+\sqrt{5}} \right) - a(\sqrt{5}-\sqrt{2})$$

$$= - \left[a \ln \frac{2+\sqrt{2}}{1+\sqrt{5}} + a(\sqrt{5}-\sqrt{2}) \right] \quad \text{we take absolute value}$$

$$\therefore S = + \left[a \ln \frac{2+\sqrt{2}}{1+\sqrt{5}} + a(\sqrt{5}-\sqrt{2}) \right]$$

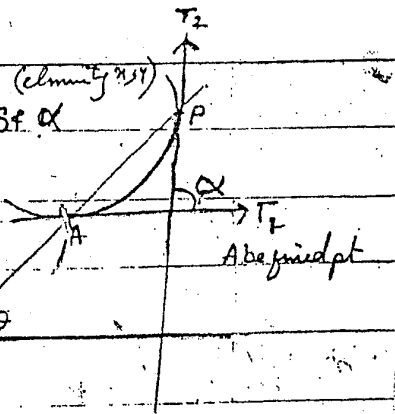
7. 6-23

V. Intrinsic

Intrinsic Eq of a Curve is the relation between S & α

where $S =$ the Arc length measured from some fixed pt A on the curve to any pt $P(x, y)$

$\alpha =$ the angle through which tangent line turns from the fixed pt A to general pt $P(x, y)$



Working Rule $\tan \alpha = \frac{dy}{dx}$ ① $S = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$ ② Eliminate x, y from ① & ②

Note: We usually choose the fixed pt A on the curve such that tangent line at A becomes // to x-axis

Imp

Q16. Intrinsic Eq of Catenary $y = c \cosh \frac{x}{c}$ is $S = c \tan \alpha$

$$y = c \cosh \frac{x}{c}$$

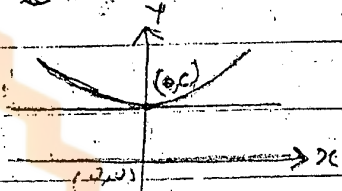
$$\frac{dy}{dx} = \sinh \frac{x}{c}$$

$$\frac{dy}{dx} = \sinh \frac{x}{c}$$

$$\text{But } \frac{dy}{dx} = \tan \alpha$$

$$\therefore \sinh \frac{x}{c} = \tan \alpha$$

Taking the vertex $(0, c)$ as fixed pt.



$$S = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_0^x \sqrt{1 + \sinh^2 \frac{x}{c}} dx = \int_0^x \cosh \frac{x}{c} dx$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$= \left[\sinh \frac{x}{c} \right]_0^x = \left(c \sinh \frac{x}{c} - c \sinh \frac{0}{c} \right)$$

$$\sinh(0) = 0$$

$$S = c \sinh \frac{x}{c}$$

$$S = c \tan \alpha$$

Proved

7.6-24i

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(Q No 17)

$$x^2 = 4ay$$

$$\frac{x^2}{4a} = y$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$

$$\text{Also } \tan \alpha = \frac{dy}{dx} = \frac{x}{2a} \therefore \frac{x}{2a} = \tan \alpha$$

$$x = 2a \tan \alpha$$

$$S' = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \int \sqrt{1 + \frac{x^2}{4a^2}} dx = \int \frac{\sqrt{4a^2 + x^2}}{2a} dx = \frac{1}{2a} \int \sqrt{(2a)^2 + x^2} dx$$

$$= \frac{1}{2a} \left[\frac{x \sqrt{(2a)^2 + x^2}}{2} + \frac{(2a)^2}{2} \ln \left(\frac{x + \sqrt{(2a)^2 + x^2}}{2a} \right) \right]_0^x$$

$$= \frac{1}{2a} \left[\frac{x \sqrt{4a^2 + x^2}}{2} + \frac{4a^2}{2} \ln \left(\frac{x + \sqrt{4a^2 + x^2}}{2a} \right) \right] - 0 - \frac{4a^2}{2} \ln \left(\frac{2a}{2a} \right)$$

$$S' = \frac{x \sqrt{4a^2 + x^2}}{4a} + a \ln \left(\frac{x + \sqrt{4a^2 + x^2}}{2a} \right)$$

$$S' = \frac{2a \tan \alpha}{4a} \sqrt{4a^2 + 4a^2 \tan^2 \alpha} + a \ln \left(\frac{2a \tan \alpha + \sqrt{4a^2 + 4a^2 \tan^2 \alpha}}{2a} \right)$$

$$= \frac{\tan \alpha}{2} \sqrt{4a^2 (1 + \tan^2 \alpha)} + a \ln \left(\frac{2a \tan \alpha + \sqrt{4a^2 (1 + \tan^2 \alpha)}}{2a} \right)$$

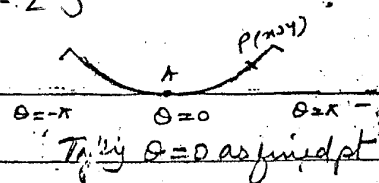
$$= \frac{\tan \alpha}{2} \cdot 2a \sec \alpha + a \ln \left(\frac{2a \tan \alpha + 2a \sqrt{\sec^2 \alpha}}{2a} \right)$$

$$= \tan \alpha \cdot a \sec \alpha + a \ln \left(\frac{2a (\tan \alpha + \sec \alpha)}{2a} \right)$$

$$= a \tan \alpha \sec \alpha + a \ln (\tan \alpha + \sec \alpha)$$

proved

7.6-25



Q19. $x = a(\theta + \sin \theta)$ $y = a(1 - \cos \theta)$

Take $\theta = 0$ as fixed pt

$$\frac{dx}{d\theta} = a(1 + \cos \theta) \quad \frac{dy}{d\theta} = a \sin \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \frac{d\theta}{dx} = \frac{a \sin \theta}{a(1 + \cos \theta)} = \frac{\sin \theta}{1 + \cos \theta}$$

$$S = \int_0^\theta \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

$$= \frac{a(1 + \cos \theta)}{2 \cos^2 \frac{\theta}{2}} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = \tan \frac{\theta}{2}$$

$$= \int_0^\theta \sqrt{a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta} d\theta$$

$$\frac{dy}{dx} = \tan \frac{\theta}{2}$$

$$\text{Also } \frac{dy}{dx} = \tan \alpha$$

$$\therefore \boxed{\alpha = \frac{\theta}{2}} \quad \text{--- (1)}$$

$$= \int_0^\theta \sqrt{a^2[1 + \cos^2 \theta + 2 \cos \theta + \sin^2 \theta]} d\theta$$

$$= \int_0^\theta \sqrt{a^2(1 + \cos^2 \theta + \sin^2 \theta + 2 \cos \theta)} d\theta$$

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$$= \int_0^\theta \sqrt{a^2(2 + 2 \cos \theta)} d\theta = \int_0^\theta \sqrt{a^2 2(1 + \cos \theta)} d\theta$$

$$= \int_0^\theta \sqrt{2a^2(2 \cos^2 \frac{\theta}{2})} d\theta = \int_0^\theta 2a \cos \frac{\theta}{2} d\theta$$

$$= 2a \left| \frac{\sin \frac{\theta}{2}}{\frac{1}{2}} \right|_0^\theta = 2a \left(2 \sin \frac{\theta}{2} - 0 \right) = 4a \sin \frac{\theta}{2}$$

Put (1) in (11)

$$S = 4a \sin \frac{\theta}{2}$$

$$S = 4a \sin \alpha$$

Ans

7.6-26

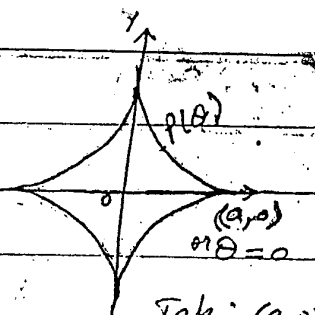
Imp

$$18 \quad x^{2/3} + y^{2/3} = a^{2/3}$$

Parametric Eqs are $\left. \begin{aligned} x &= a \cos^3 \theta \\ y &= a \sin^3 \theta \end{aligned} \right\} \quad \begin{aligned} \frac{dx}{d\theta} &= 3a \cos^2 \theta (-\sin \theta) \\ \frac{dy}{d\theta} &= 3a \sin^2 \theta (\cos \theta) \end{aligned}$

$$\tan \alpha = \frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^3 \theta \sin \theta} = -\tan \theta$$

$$\therefore \tan \alpha = -\tan \theta = \tan(-\theta) \Rightarrow \boxed{\alpha = -\theta}$$



Taking (a, 0) as fixed pt. or $\theta = 0$ as fixed pt.

$$S = \int_0^{\theta} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta = \int_0^{\theta} \sqrt{9a^2 \cos^4 \theta \sin^2 \theta + 9a^2 \sin^4 \theta \cos^2 \theta} d\theta$$

$$= \int_0^{\theta} \sqrt{9a^2 \sin^2 \theta \cos^2 \theta (\cos^2 \theta + \sin^2 \theta)} d\theta = \int_0^{\theta} 3a \sin \theta \cos \theta d\theta$$

$$S' = \left| \frac{3a \sin^2 \theta}{2} \right|_0^{\theta} = \left(\frac{3a \sin^2 \theta}{2} - 0 \right) = \frac{3a \sin^2 \theta}{2}$$

$$S = \frac{3a \sin^2(-\alpha)}{2} = \frac{3a}{2} [\sin(-\alpha)]^2 = \frac{3a \sin^2 \alpha}{2}$$

OR

$$\frac{dy}{dx} = \frac{\sin \theta}{-\cos \theta} = -\tan \theta = \tan(\pi - \theta)$$

$$\text{but } \frac{dy}{dx} = \tan \alpha$$

$$\therefore \tan \alpha = \tan(\pi - \theta)$$

$$= \pi - \theta \Rightarrow \theta = \pi - \alpha$$

$$\therefore S' = \frac{3a \sin^2 \theta}{2} = \frac{3a \sin^2(\pi - \alpha)}{2} = \frac{3a \sin^2 \alpha}{2}$$

Ans. $\frac{3a \sin^2 \alpha}{2}$

